Slides adapted from B. Bert Gertsman, other sources are referenced on the slides

CS4048 – DATA SCIENCE

Lecture 14 & 15

October 15 & 17, 2025

Fall 2025

FAST – NUCES, CFD Campus

Dr. Rabia Maqsood

rabia.maqsood@nu.edu.pk

TODAY'S TOPICS

- Inferential statistics role in data science
- Standard/sampling error
- Confidence interval
  - Significance level
  - z-test, t-test critical value
- Hypothesis/significance testing
  - Null & alternative hypothesis
  - Critical value
  - P-value
  - z-test, t-test, Chi-square test

INFERENTIAL STATISTICS

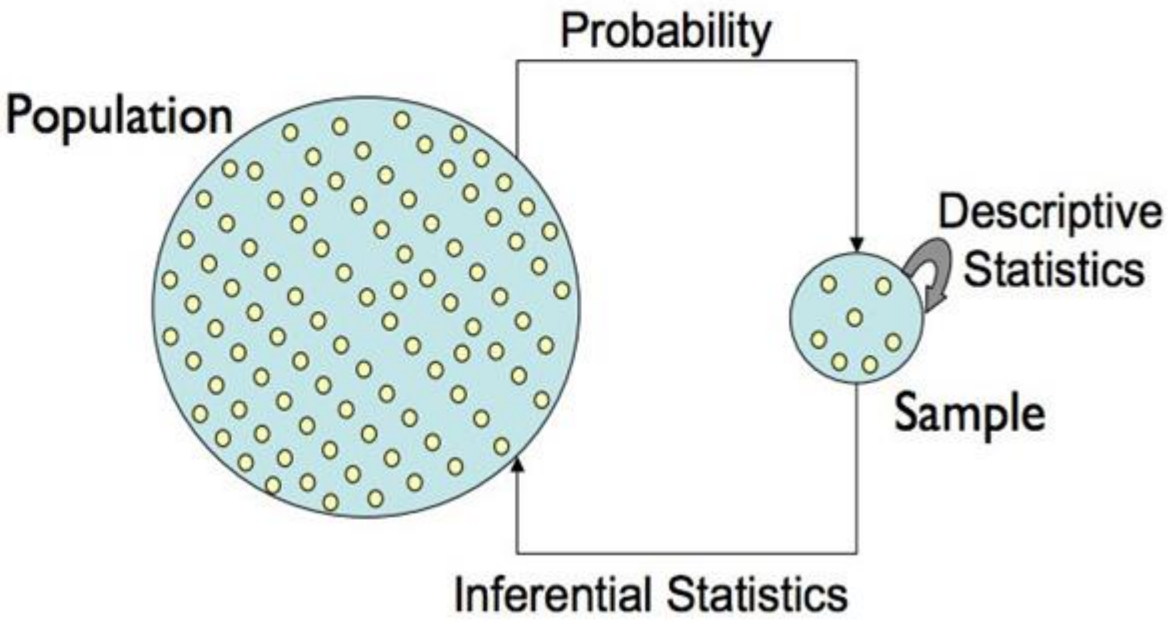
- Objective is to make inferences based on sample data collected and draw conclusions about the population.

- Examples:
  - The average expectancy of life have declined over years in country X.
  - Men and Women have different average heights in a population.
  - Medicine B has a better effect on patients health than medicine A.
  - …

- More relevant to data science:
  - What is the 95% confidence interval for predicted sales next month?
  - Is the mean squared error (MSE) improvement statistically significant?
  - Is the trend in monthly sales significant?
  - Does classifier A perform significantly better than classifier B?
  - Does showing recommendations increase average order value?
  - …

INFERENTIAL STATISTICS

- **Point estimate:** a single value that is used to estimate a population parameter
  - Example: estimating population mean (salary) based on a sample data
  - In data science, point estimate could be a model's accuracy measure (in classification).



SAMPLING ERRORS

- Sampling error includes:

1. **Systematic sampling error:** introduced by errors in data collection or data not draw correctly (sampling bias), thus, the sample data is not a true representation of the population.
2. **Random sampling error (or chance error or standard error):** variability in results occur due to a random chance
   - Samples drawn from the same population will rarely provide identical estimates of the population parameter of interest

- Systematic sampling error is the fault of the investigation, but random sampling error is not.

STANDARD ERROR

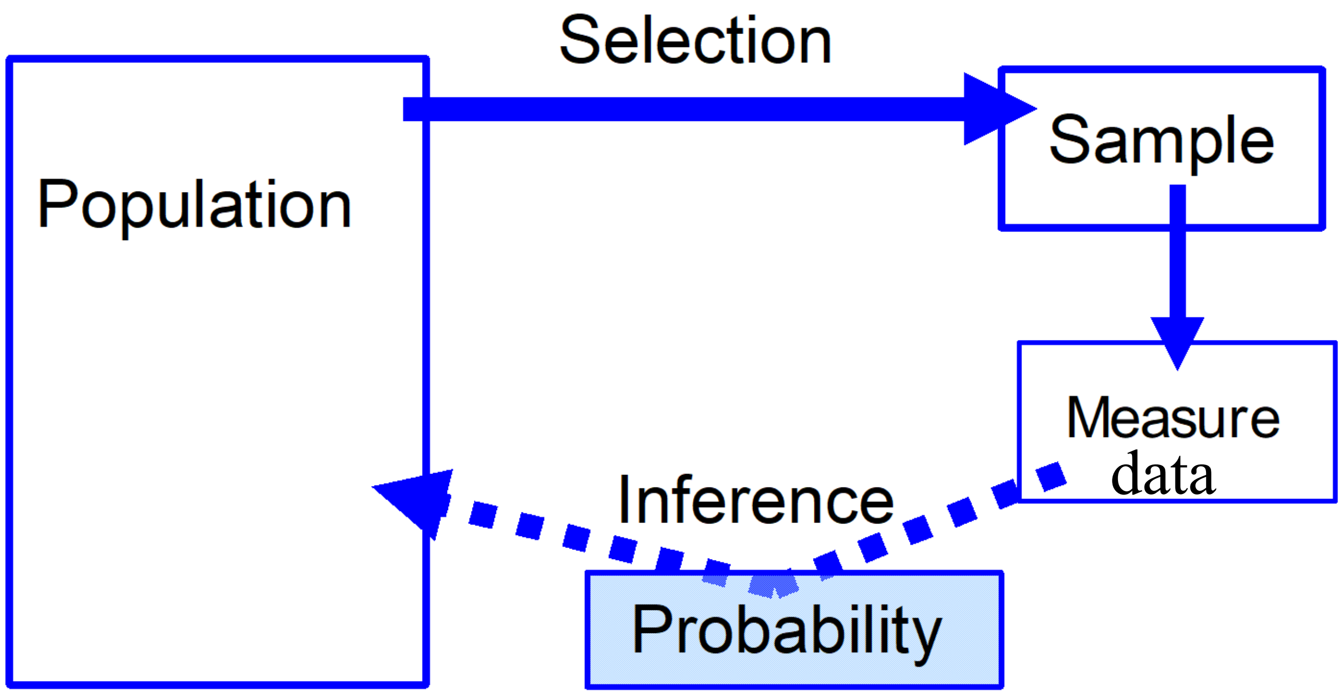
- Standard error is a single metric that sums up the variability in the sampling distribution of a statistic.

- Even with the best sampling techniques, some degree of random error is expected.

- Standard error can be represented statistically to support inference results interpretation.
  - S.E $= \frac{\sigma}{\sqrt{n}}$

- **Statistical probability:** the probability associated with a statistic is the level of confidence that we have in an estimated value

Chain of Reasoning for Inferential Statistics

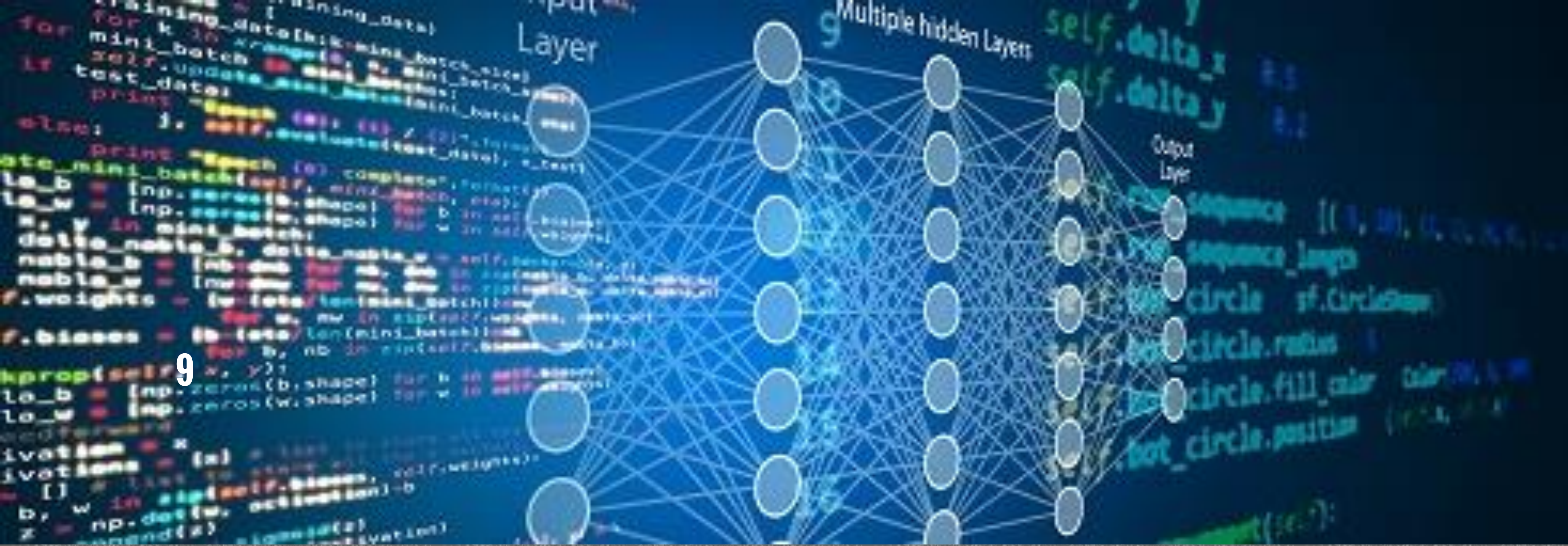


Are our inferences valid?…Best we can do is to calculate probability about inferences

INFERENTIAL STATISTICS

- There are two forms of statistical inference:

  1. Confidence intervals
  2. Hypothesis ("significance") tests



CONFIDENCE INTERVAL

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CONFIDENCE INTERVAL (CI)

Two forms of estimation

- **Point estimation** ≡ most likely value of parameter (e.g., x-bar is point estimator of μ)
- **Interval estimation** ≡ range of values with known likelihood of capturing the parameter, i.e., a **confidence interval (CI)**

CONFIDENCE INTERVAL (CI)

- Confidence interval provides a level of uncertainty associated with the estimate and is expressed as a range of values.

- To calculate confidence interval, two elements are needed:
  1. A statistic (or point estimate)
  2. Level of confidence
  3. Margin of error (denoted by E)

- Compute the confidence interval (CI) as:

$$\text{Lower Bound of the Confidence Interval} = \text{Point Estimate} - \text{Margin of Error}$$
$$\text{Upper Bound of the Confidence Interval} = \text{Point Estimate} + \text{Margin of Error}$$

MARGIN OF ERROR (E)

- **Margin of error:** provides an indication of the maximum error of the point estimate (or test statistic)
  - A lower margin of error will increase our confidence in the point estimate
  - A higher margin of error will decrease our confidence in the point estimate

  - **Margin of error (E) = Critical value * Standard error**

Obtained from a data distribution,
e.g., z or t distribution

$$\frac{\sigma}{\sqrt{n}}$$

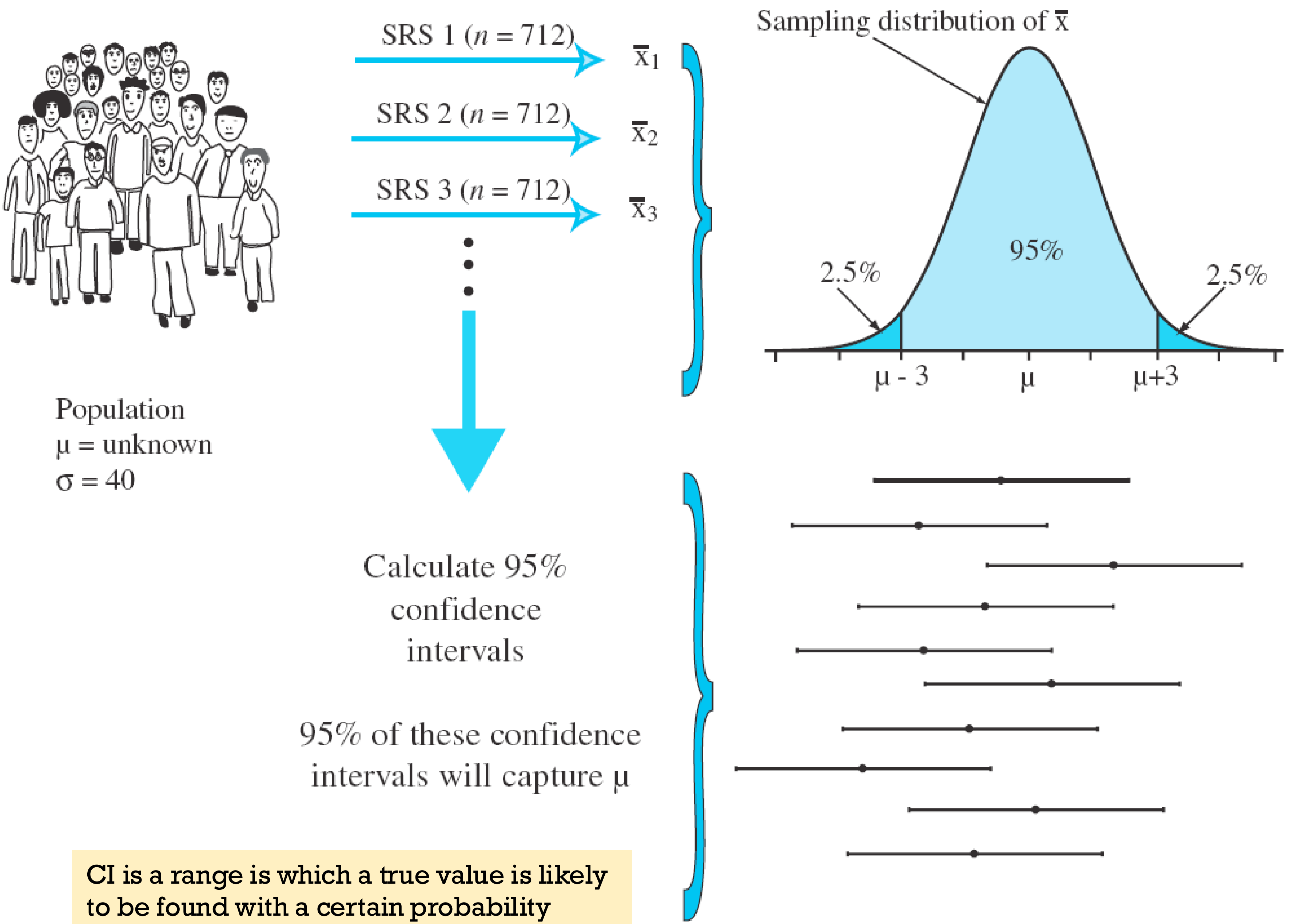
CONFIDENC LEVEL

- In other words, the margin of error reflects the desired ***confidence level*** of the researcher.

- The confidence level is the **probability** that the interval estimate will contain the population parameter, given that the estimation process on the parameter is repeated over and over.
  - Example: A 95% confidence level means that if you were to take 100 different samples and create a 95% confidence interval (CI) for each one, you would expect 95 of those CIs to contain the true population value.

- CI is a range is which a true value is likely to be found with a certain probability.
- CI = $(1.0 - \alpha)$, usually expressed as a percentage (e.g. 95%).

- For probability, we define a high confidence level usually 95% or 99% (or sometimes 90%).



CI is a range is which a true value is likely to be found with a certain probability

RESULTS INTERPRETATION

- Consider a data scientist who is interested in forecasting the median income for all residents of California. Income data is collected from a sample of 1,000 residents, and the median income level is $68,500.
- Also assume the margin of error for a 95% confidence interval is $4,500.
- Now the data scientist can construct a 95% confidence interval to forecast income levels as follows:

$$\begin{aligned}
\text{Lower Bound of the Confidence Interval} &= \text{Point Estimate} - \text{Margin of Error} \\
&= \$68500 - \$4500 \\
\text{Upper Bound of the Confidence Interval} &= \text{Point Estimate} + \text{Margin of Error} \\
&= \$68500 + \$4500 = \$73000
\end{aligned}$$

- The data scientist can then state the following conclusion:
  - There is 95% confidence that the forecast for median income for all residents of California is between $64,000 and $73,000.

"BODY WEIGHT" EXAMPLE

- Body weights of 20-29-year-old males have unknown μ and $\sigma = 40$
- Take an SRS of $n = 712$ from population
- Calculate: x-bar =183

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{712}} = 1.5 \text{ and } m \approx 2 \times SE_{\bar{x}} = 2 \times 1.5 = 3$$

$$\begin{aligned} 95\% \text{ CI for } \mu &\approx \bar{x} \pm m \\ &= 183 \pm 3 \\ &= 180 \text{ to } 186 \text{ pounds} \end{aligned}$$

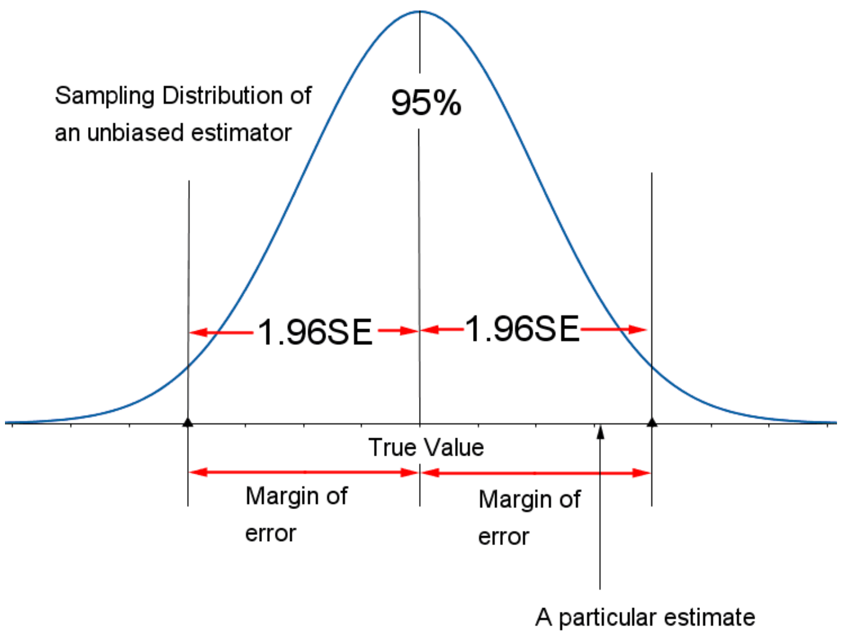
CONFIDENCE INTERVAL FORMULA

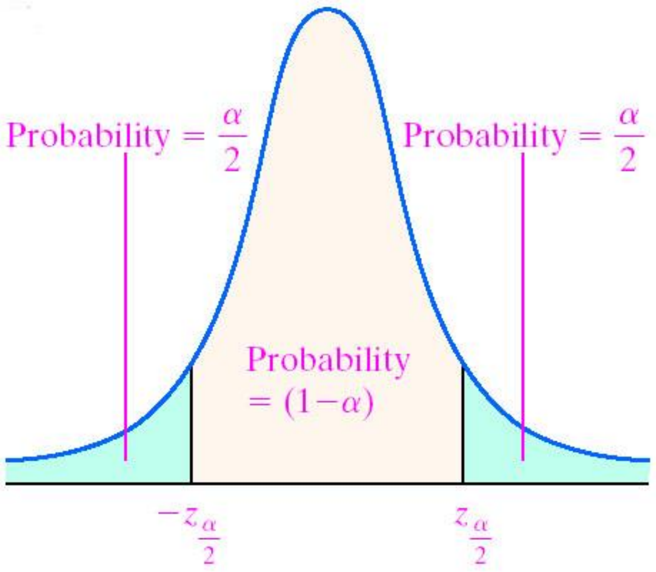
Here is a more accurate and flexible formula

Critical value

$$\bar{x} \pm z_{1-\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

$$\text{Equivalently, } \bar{x} \pm z_{1-\frac{\alpha}{2}} \cdot SE_{\bar{x}}$$





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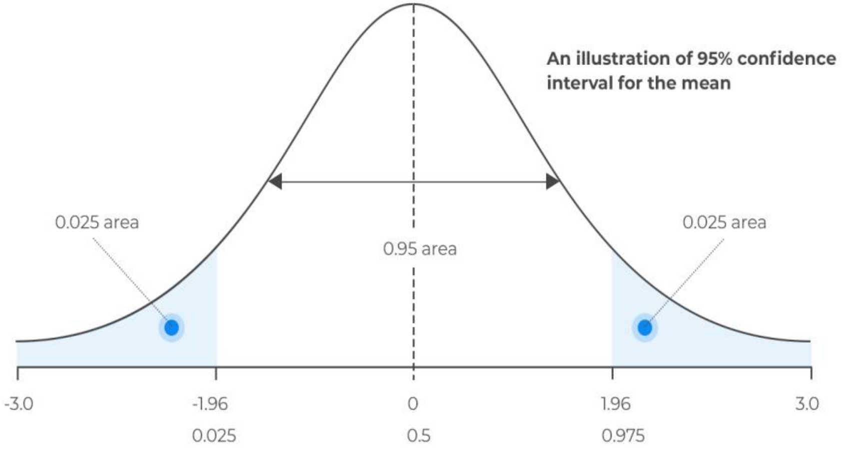
Z TABLE



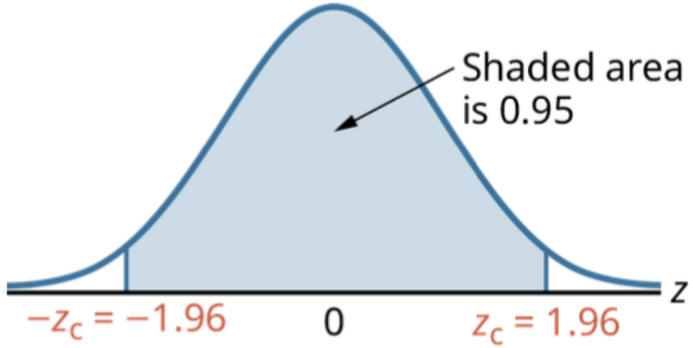


Figure 4.2 Critical Values of z_c Marked on Standard Normal Curve for 95% Confidence Interval

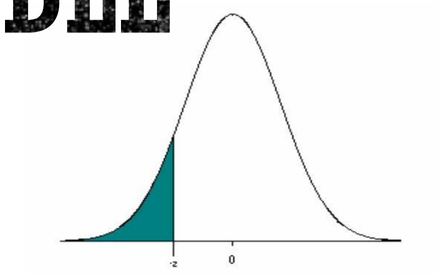
Calculate Z-score for 95% confidence interval

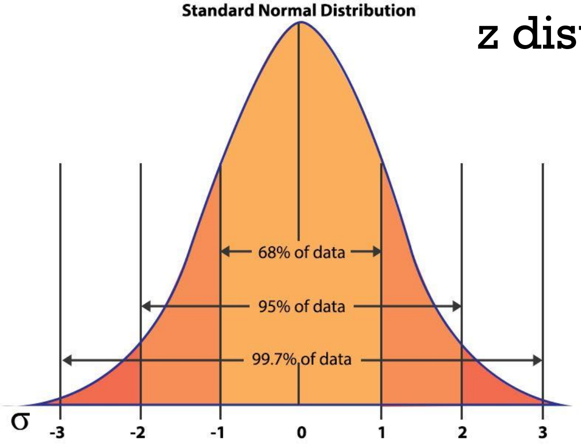
Z-score	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

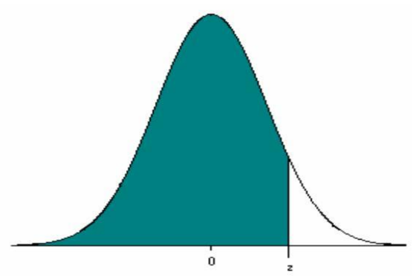
MATHBLOG

Z TABLE

z distribution a.k.a "standard normal distribution"
Mean = 0, SD = 1







z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
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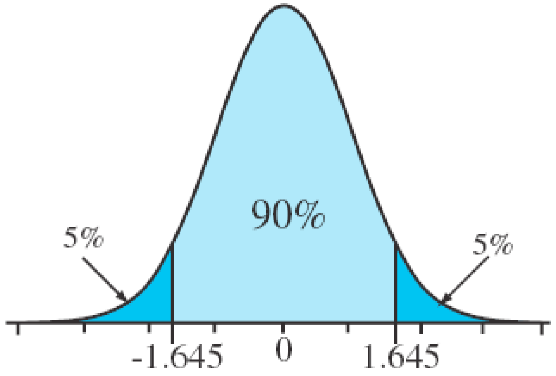
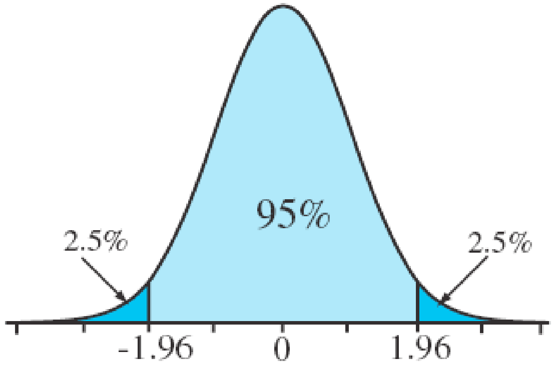
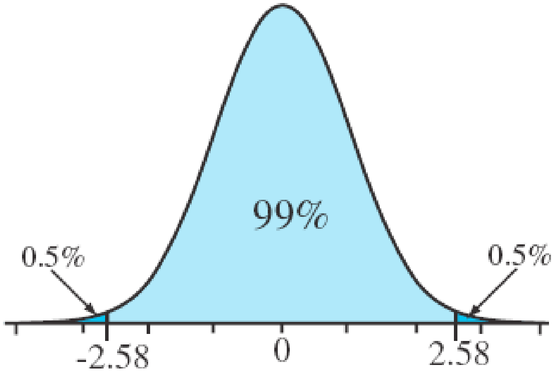
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

COMMON LEVELS OF CONFIDENCE

Confidence level $1 - \alpha$	Alpha level α	Z value $z_{1-(\alpha/2)}$
.90	.10	1.645
.95	.05	1.960
.99	.01	2.576







90% CONFIDENCE INTERVAL FOR Μ

Data: SRS, n = 712, σ = 40, x-bar = 183

$$90\%\ \text{CI for } \mu = \bar{x} \pm z_{1-\frac{.1}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

$$= 183 \pm 1.645 \cdot \frac{40}{\sqrt{712}}$$

$$= 183 \pm 2.5$$

$$= 180.5 \text{ to } 185.5$$

95% Confidence Interval for μ

Data: SRS, n = 712, σ = 40, x-bar = 183

$$95\%\ \text{CI for}\ \mu = \bar{x} \pm z_{1-\frac{.05}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

$$= 183 \pm 1.960 \cdot \frac{40}{\sqrt{712}}$$

$$= 183 \pm 2.9$$

$$= 180.1\ \text{to}\ 185.9$$

99% CONFIDENCE INTERVAL FOR Μ

Data: SRS, n = 712, σ = 40, x-bar = 183

$$99\%\ \text{CI for}\ \mu = \bar{x} \pm z_{1-\frac{.01}{2}} \cdot \frac{\sigma}{\sqrt{n}}$$

$$= 183 \pm 2.576 \cdot \frac{40}{\sqrt{712}}$$

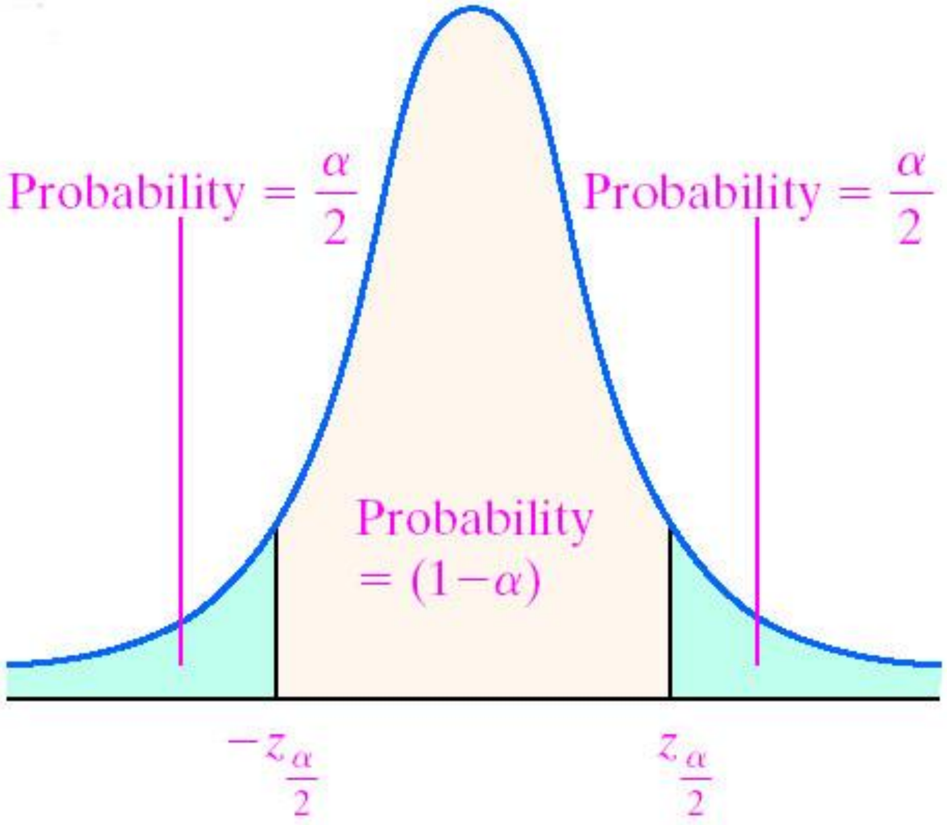
$$= 183 \pm 3.9$$

$$= 179.1\ \text{to}\ 186.9$$

Z-DISTRIBUTION

- Use z-distribution when populatin SD is known and sample size >= 30
- Margin of erro:

$$E = z_c \frac{\sigma}{\sqrt{n}}$$



z_c is called the critical value of the normal distribution and is calculated as the z-score, which includes the ar corresponding to the confidence level between $-z_c$ and $+z_c$. For example, for a 95% confidence interval, the corresponding critical value is $z_c = 1.96$ since an area of 0.95 under the normal curve is contained between the z-scores of -1.96 and $+1.96$.

σ is the population standard deviation.

n is the sample size.

T-DISTRIBUTION

- Use **t-distribution** when populatin SD is unknown and smaller sample size < 30

(t-distribution is based on degrees of freedom, df = n-1)

- Margin of error (E) is calculated as:

$$E = t_c \frac{s}{\sqrt{n}}$$

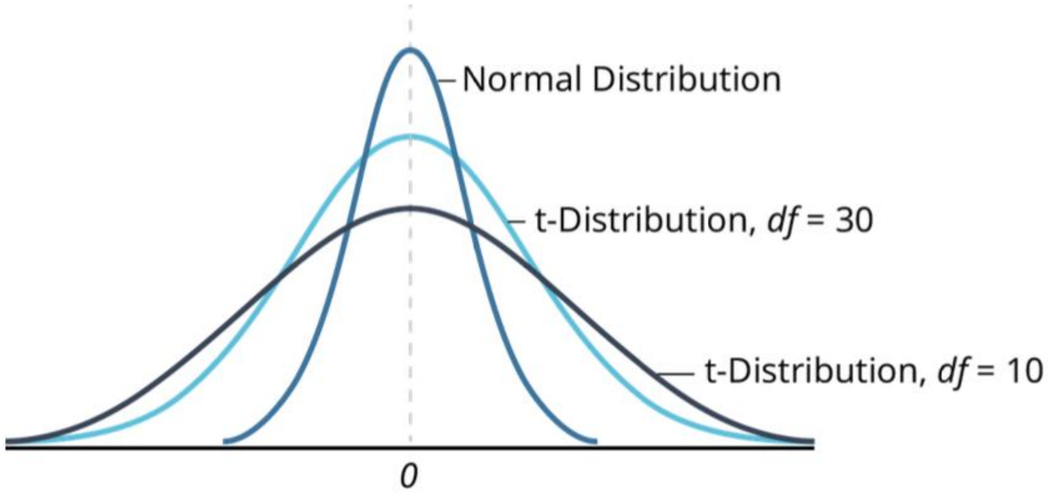
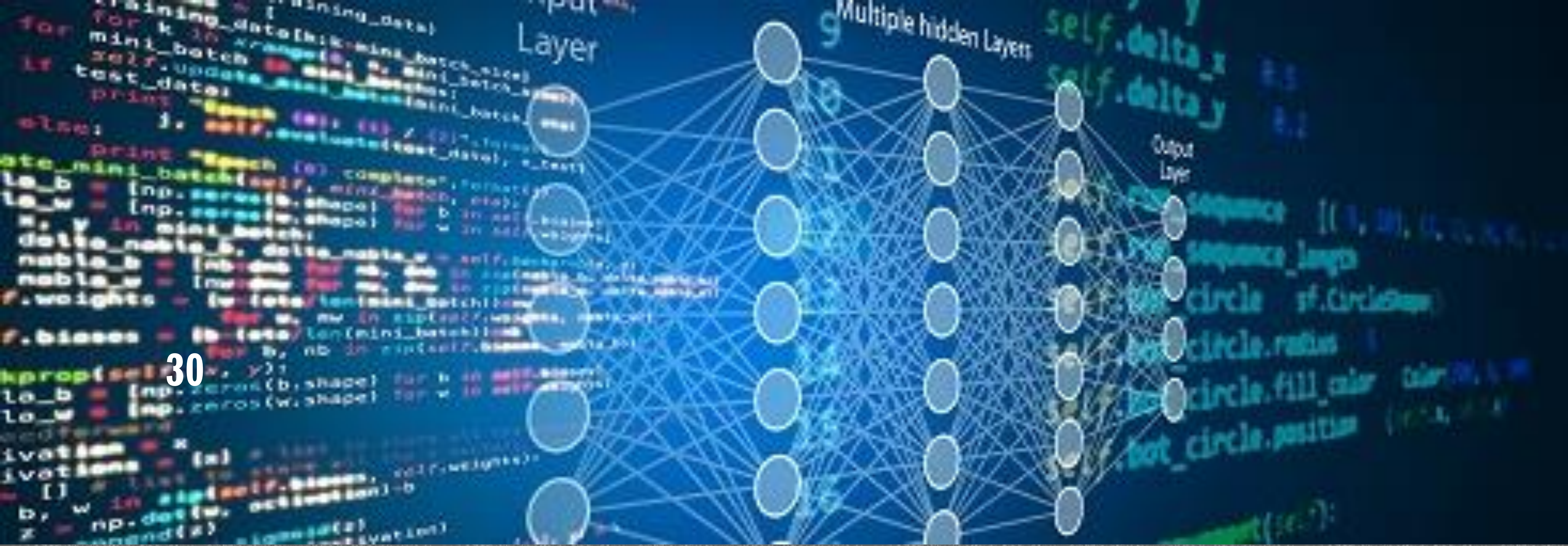


Figure 4.3 Comparison of t-Distribution and Normal Distribution Curves

T TABLE

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										



HYPOTHESIS TESTING

Inferential Statistics: uses sample data to evaluate the credibility of a hypothesis about a population

↓

NULL Hypothesis:

NULL (*nullus* - latin): "not any" → no differences between means

$$H_0 : \mu_1 = \mu_2$$

"H- Naught"

Always testing the null hypothesis

Inferential statistics: uses sample data to evaluate the credibility of a hypothesis about a population

$\downarrow$

Hypothesis: Scientific or alternative hypothesis

Predicts that there are differences between the groups

$$H_1 : \mu_1 \neq \mu_2$$

Hypothesis

A statement about what findings are expected

null hypothesis

"the two groups will not differ"

alternative hypothesis

"group A will do better than group B"
"group A and B will not perform the same"

Alternative hypothesis is typically the statement that the researcher would like to prove or verify (i.e., researcher's claim).

EXAMPLES

- Null = “no difference between the means of group A and group B”

Alternative = “A is different from B” (could be bigger or smaller)

- Null = “A <= B”

Alternative = “A > B”

- Null = “B is not X% greater than A”

Alternative = “B is X% greater than A”

WHY HYPOTHESIS TESTING?

- Why to perform a significance/hypothesis test when we can just compare the results and make a conclusion?

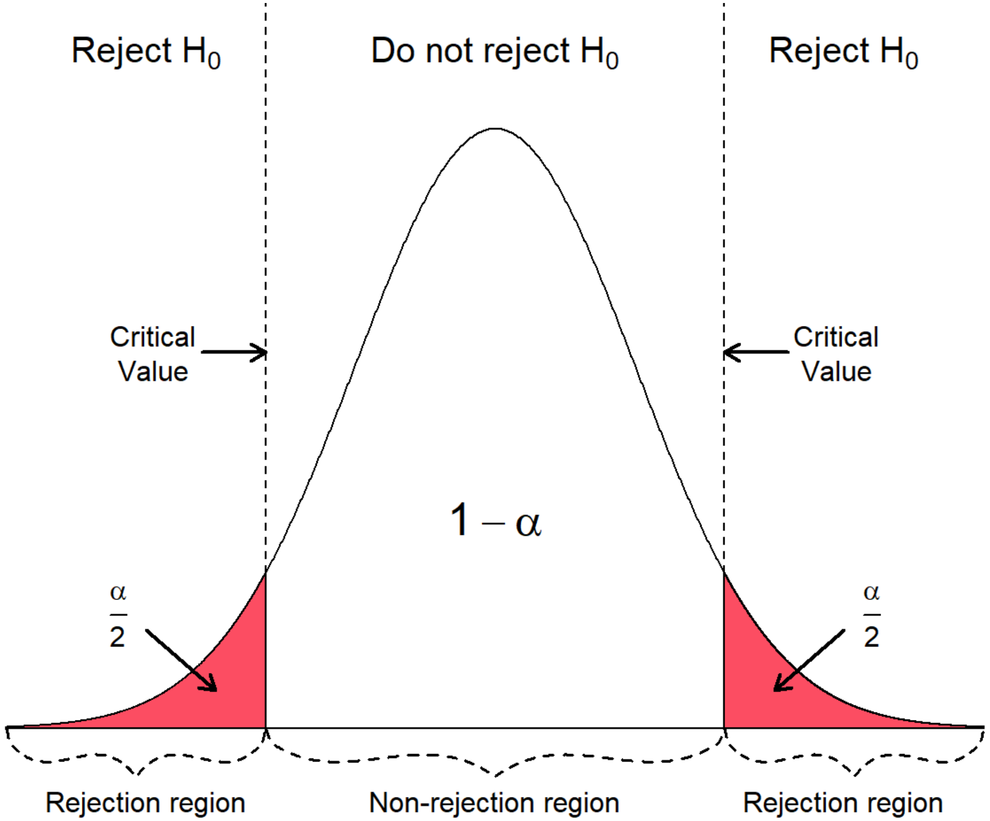
- For any single experiment, we are bound to see a difference, just as we see a difference between the means of two random samples in a distribution of sample means.

- If the null hypothesis is true, then differences in mean scores are just two random samples from the same population.

- <u>Testing the null hypothesis:</u> **A statistical test assesses the probability of obtaining a given sample or samples of scores, assuming the null hypothesis is correct.**

HYPTHESIS TESTING

If Null hypothesis is rejected, then we say that the result is **statistically significant,** for some level of significance(alpha).



Ref: https://www.geo.fu-berlin.de/en/v/soga-r/Basics-of-statistics/Hypothesis-Tests/Introduction-to-Hypothesis-Testing/Critical-Value-and-the-p-Value-Approach/index.html

Inferential Statistics

When making comparisons btw 2 sample means there are 2 possibilities

Null hypothesis is true

Not reject the Null Hypothesis

Null hypothesis is false

Reject the Null hypothesis

Possible Outcomes in Hypothesis Testing (Decision)

	Null is True	Null is False
Accept	Correct Decision	Error Type II Error
Reject	Error Type I Error	Correct Decision

Type II error: mistakenly concluding an effect is due to chance (when it is real)

Type I error: mistakenly concluding an effect is real (when it is due to chance)

Type I Error: Rejecting a True Hypothesis
Type II Error: Accepting a False Hypothesis

Which of these two types of error is more serious??

TYPE I & II ERRORS

- Recall that hypothesis testing is performed to guard us in making a decision against a random chance.

- Thus, significance/hypothesis tests are typically structured to minimize Type I errors.

- Type II error is not so much an error as a judgment that the sample size is too small to detect the effect (i.e., a larger sample might yield the true effect).

Hypothesis Testing - Decision

Decision Right or Wrong?

But we can know the probability of being right or wrong

Can specify and control the probability of making TYPE I of TYPE II Error

Try to keep it small…

ALPHA

the probability of making a type I error → depends on the criterion you use to accept or reject the null hypothesis = significance level (smaller you make alpha, the less likely you are to commit error) **0.05** (5 chances in 100 that the difference observed was really due to sampling error – 5% of the time a type I error will occur)

Possible Outcomes in Hypothesis Testing

Alpha (α)

Difference observed is really just sampling error
The prob. of type one error

	Null is True	Null is False
Accept	Correct Decision	Error Type II Error
Reject	Error Type I Error	Correct Decision

WHEN NULL HYPOTHESIS IS TRUE (TYPE I ERROR)

- If null hypothesis is true and alpha level is set at .05, then the null hypothesis will be rejected 5% of time even though it is true.

- One way to safeguard against a Type I error is to set a more stringent alpha level (e.g., $p<.01$).

SETTING THE PROBABILITY OF A TYPE I ERROR

- In trying to minimize, however, the probability of a Type I error, we encounter an obstacle in that the probabilities of the Type I and Type II errors are inversely related. Thus, if we try to make the probability of a Type I error very, very small, then it will make the probability of a Type II error quite large.

- As a compromise we therefore specify a maximum tolerable Type I error probability, called the significance level, and denoted by α, and choose the critical value C such that the probability of a Type I error is (at most) equal to α.
- This α is conventionally set to 0.10, 0.05, or 0.01.

BETA

Probability of making type II error → occurs when we fail to reject the Null when we should have

Possible Outcomes in Hypothesis Testing

Beta (β)

Difference observed is real
Failed to reject the Null

	Null is True	Null is False
Accept	Correct Decision	Error Type II Error
Reject	Error Type I Error	Correct Decision

POWER: ability to reduce type II error

POWER: ability to reduce type II error
(1-Beta) – Power Analysis

The power to find an effect if an effect is present

1. Increase our n

2. Decrease variability

3. More precise measurements

Effect Size: measure of the size of the difference between means attributed to the treatment

HYPOTHESIS TESTING STEPS

- Hypothesis tests involve the following steps:

1. State the null and alternative hypotheses
2. Select a level of significance
3. Choose a test criterion (z-test, t-test, chi-square, etc.) & state the decision rule
4. Calculate the test statistic
5. Make conclusion (reject or fail to reject the null hypothesis)

1. STATE HYPOTHESES: ONE WAY TEST

- One-way test: hypothesis test that counts chance results only in one direction.
  - One-tailed test predicts the direction of the effect or relationship

  - Examples:

1. RQ1: Does the new sales strategy increase average monthly revenue?
   - Null hypothesis (H_0): The new strategy does not increase revenue. ($\mu_1 \leq \mu_2$)
   - Alternative hypothesis (H_1): The new strategy increases revenue. ($\mu_1 > \mu_2$)

2. RQ2: Does the new deep learning model achieve higher accuracy than the baseline model?
   - H_0: Accuracy(new) $\leq$ Accuracy(baseline)
   - H_1: Accuracy(new) > Accuracy(baseline)

1. STATE HYPOTHESES: TWO WAY TESTING

- Two-way test: hypothesis test that counts chance results in two directions.
  - A.k.a two tailed hypothesis

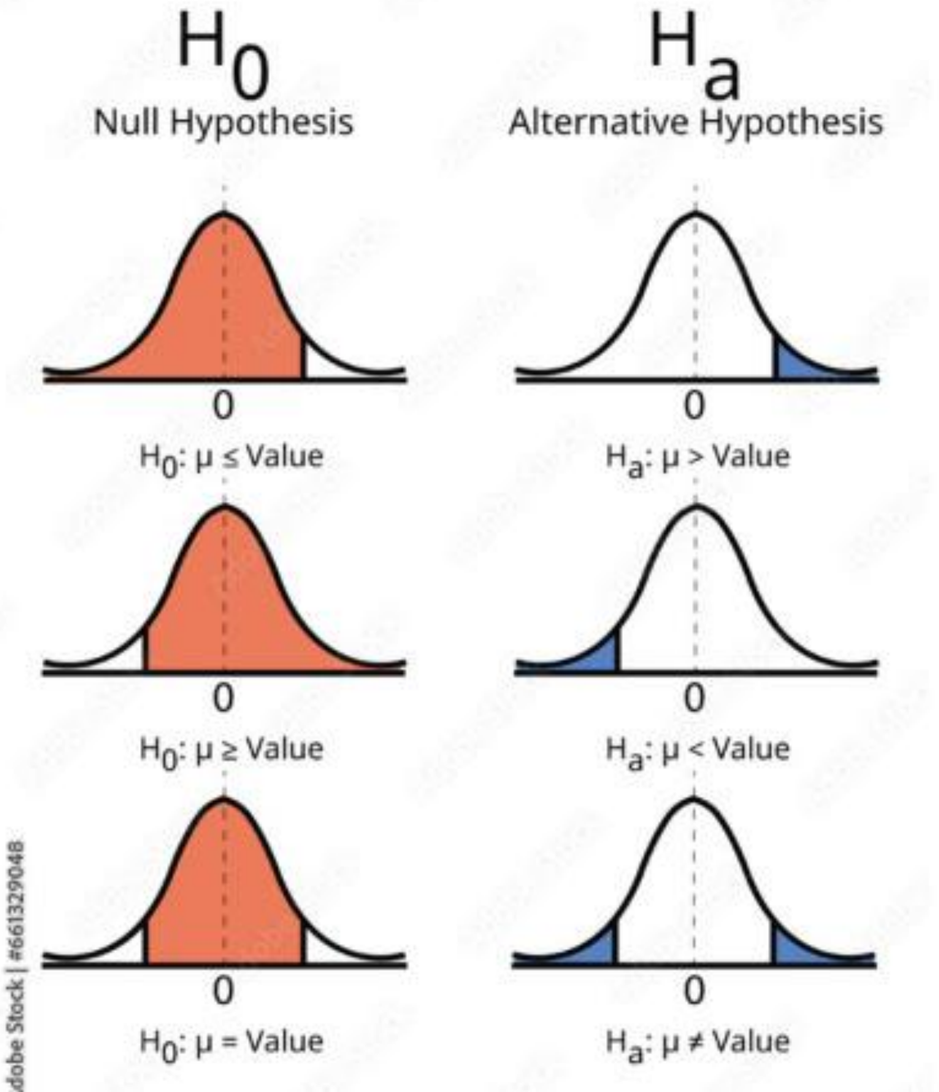
  - Examples:

1. RQ1: Do male and female students differ in their average exam scores?
   - H_0: The average exam scores are same for both genders ($\mu_1 = \mu_2$)
   - H_1: The average exam scores are not same for both genders ($\mu_1 \neq \mu_2$)

2. RQ2: Does changing the website layout affect conversion rate (in either direction)? *(A/B testing)*
   - H_0: Conversion_A = Conversion_B
   - H_1: Conversion_A ≠ Conversion_B

Generally, two-tailed tests are preferred to one-tailed tests.

ONE VS TWO TAILED TEST



HYPOTHESIS TESTING STEPS

- Hypothesis tests involve the following steps:

1. State the null and alternative hypotheses
2. Select a level of significance
3. Choose a test criterion (z-test, t-test, chi-square, etc.) & state the decision rule
4. Calculate the test statistic
5. Make conclusion (reject or fail to reject the null hypothesis)

2. LEVEL OF SIGNIFICANCE (ALPHA)

- In hypothesis testing, **significance** refers to when a result is very unlikely to have occurred if the null hypothesis is correct (type I or False Positive error).

- Alpha is the highest value of p that we are willing to tolerate and still say that a difference, change, or effect observed in the sample is "Statistically Significant"

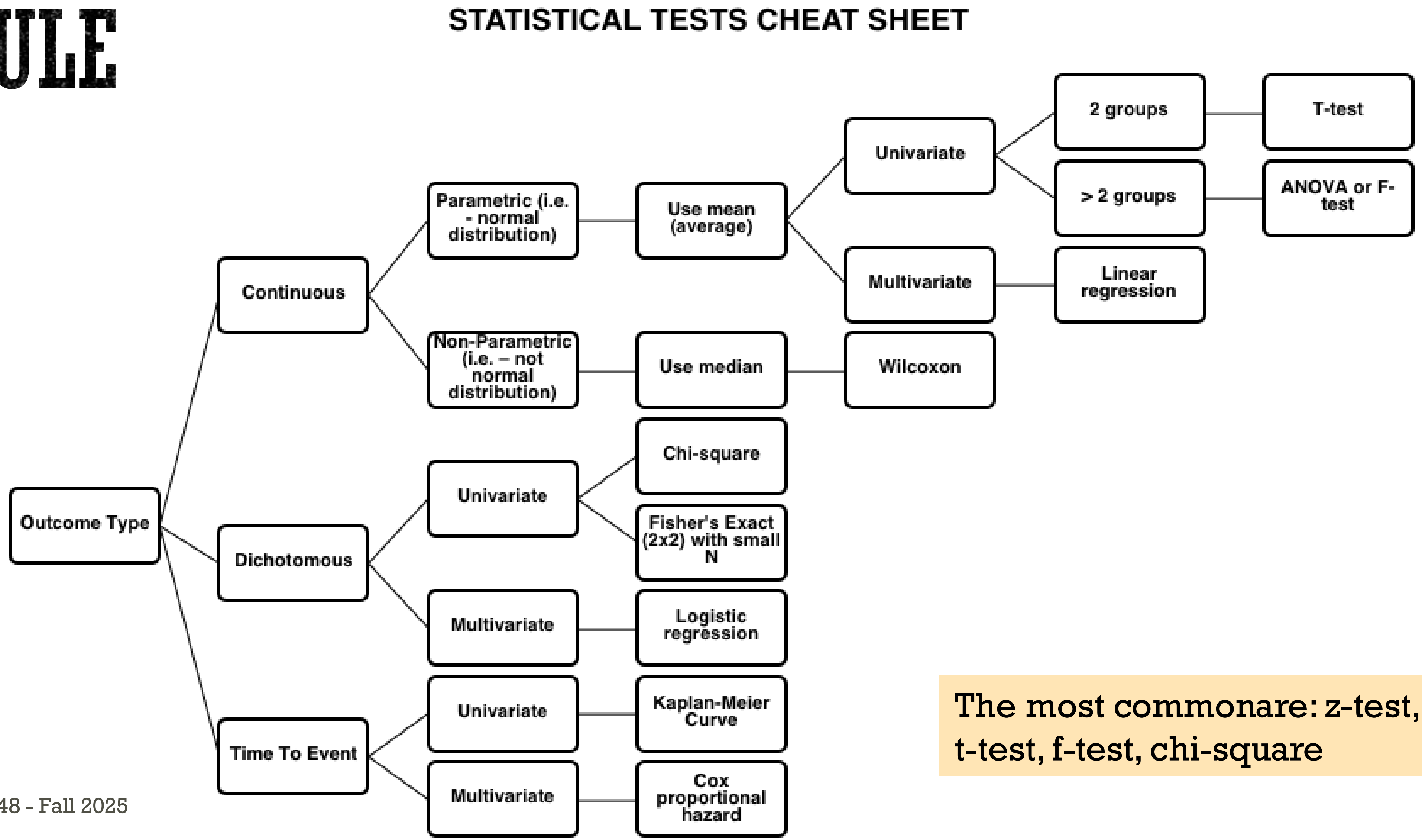
- Set the fixed **alpha level** (Type I error) to 0.05. This means, if the null hypothesis is true, the probability of incorrectly rejecting it is 5% or less.

HYPOTHESIS TESTING STEPS

- Hypothesis tests involve the following steps:

1. State the null and alternative hypotheses
2. Select a level of significance
3. Choose a test criterion (z-test, t-test, chi-square, etc.) & state the decision rule
4. Calculate the test statistic
5. Make conclusion (reject or fail to reject the null hypothesis)

3. CHOOSE TEST CRITERION & DECISION RULE



The most commonare: z-test, t-test, f-test, chi-square

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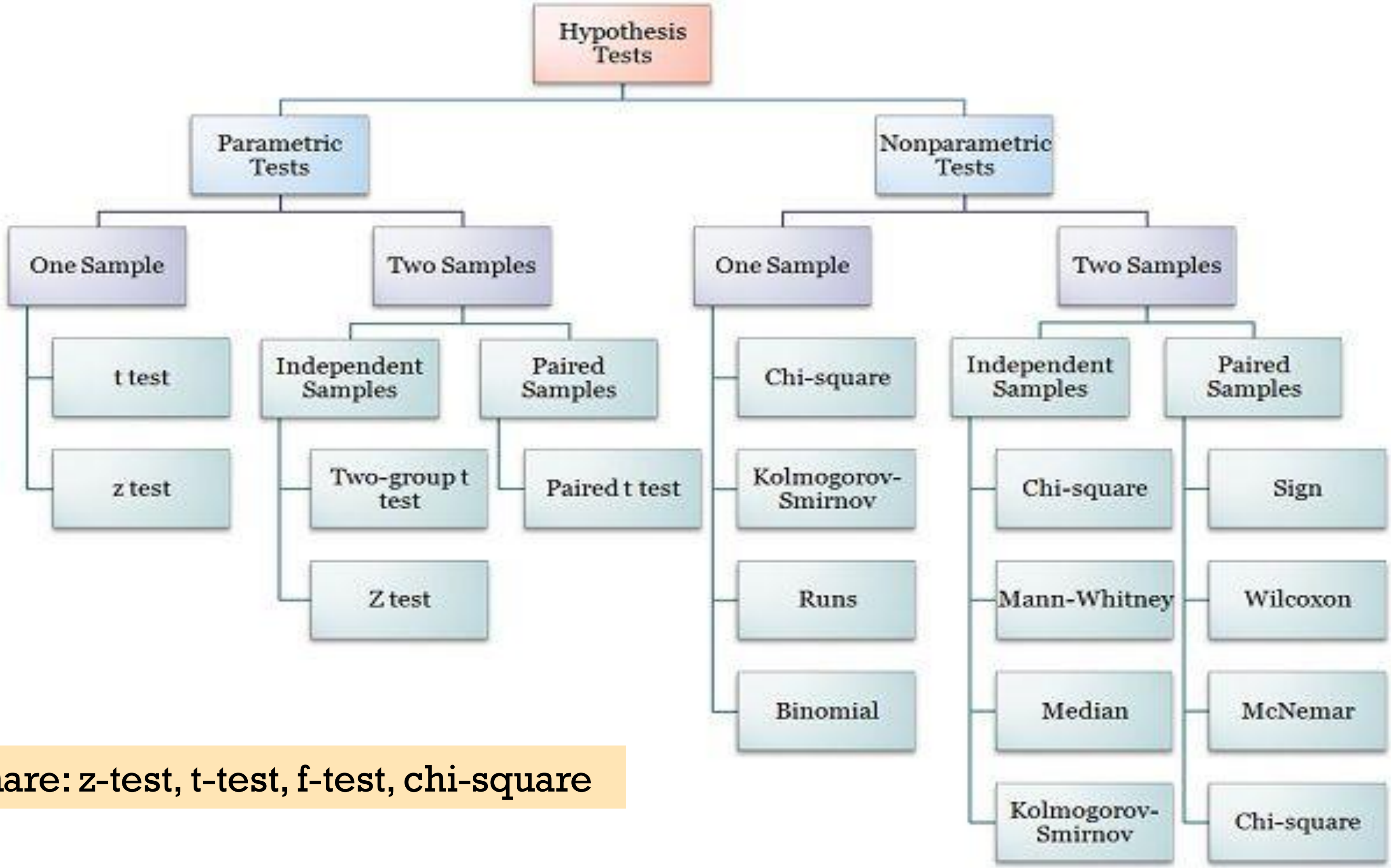
3. CHOOSE TEST CRITERION & DECISION RULE



The most commonare: z-test, t-test, f-test, chi-square

3. CHOOSE TEST CRITERION & DECISION RULE

Test statistic	Associated test	Sample size	Information given	Distribution	Test question
z-score	z-test	Two populations or large samples (n > 30)	• Standard deviation of the population (this will be given as σ) • Population mean or proportion	Normal	Do these two populations differ?
t-statistic	t-test	Two small samples (n < 30)	• Standard deviation of the sample (this will be given as *s*) • Sample mean	Normal	Do these two samples differ?
f-statistic	ANOVA	Three or more samples	• Group sizes • Group means • Group standard deviations	Normal	Do any of these three or more samples differ from each other?
chi-squared	chi-squared test	Two samples	• Number of observations for each categorical variable	Any	Are these two categorical variables independent?

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3. HYPOTHESIS TESTING DECISION

- The decision to reject or not reject the null hypothesis usually is made with reference to the sampling distribution of a statistic of some kind (e.g., z-distribution, t-distribution).

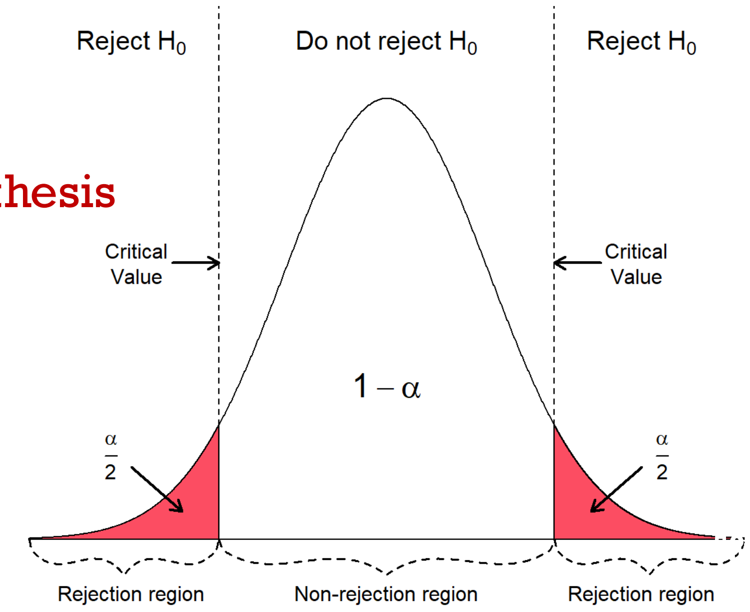
- To make a decision whether to reject the null hypothesis, a test statistic is calculated.

- There are two approaches how to arrive at that decision:
  1. The critical value approach
  2. The p-value approach

3. HYPOTHESIS TESTING DECISION USING CRITICAL VALUE

- Critical value is a cut-off value that is used to mark the start of a region where the test statistic, obtained in hypothesis testing, is unlikely to fall in.
- The critical value divides the area under the probability distribution curve in **rejection region**(s) and **non-rejection region**.

- Decision rule:
  - If the test statistic > critical value → **reject** the null hypothesis
  - If the test statistic <= critical value, **fail to reject** the null hypothesis



3. HYPOTHESIS TESTING DECISION USING CRITICAL VALUE

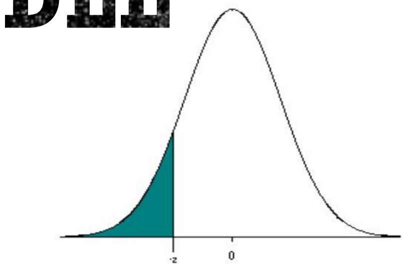
- **How to compute critical value?**
  - The critical value is computed based on the given significance level α and the type of probability distribution of the idealized (or null) model.
- We've already done this in confidence interval

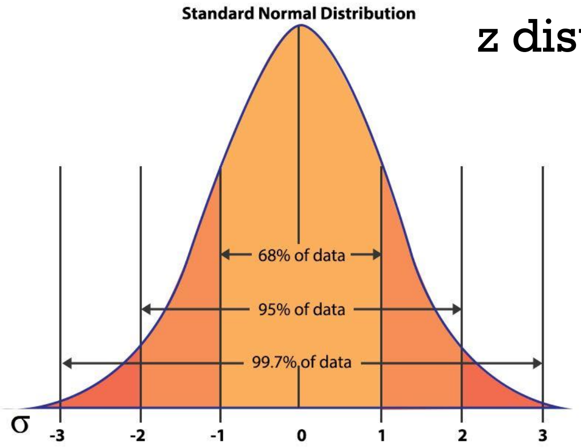
TODO: Find critical value for a two-tailed z-test with α=0.01

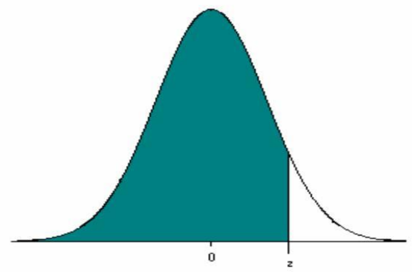
https://www.statisticsfromatoz.com/uploads/7/3/2/1/73216723/crit-value-jpg_orig.jpg

Z TABLE

z distribution a.k.a "standard normal distribution"
Mean = 0, SD = 1



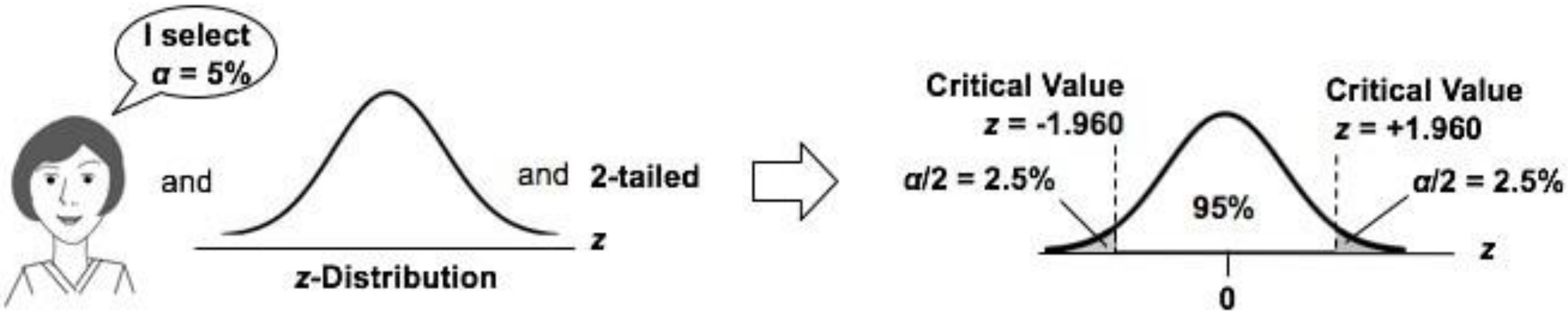




z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

3. HYPOTHESIS TESTING DECISION USING CRITICAL VALUE



To be continued…

READING MATERIAL

- Chapter#3: Principle of Data Science
- Chapter#1, 2: Practical Statistics [Bruce]
- An article: https://www.kdnuggets.com/2021/03/statistics-data-scientists-should-know.html
- A useful resoure on inferential statistics: https://www.statisticsfromatoz.com/blog
- Z-test in ML: https://mlvidhya.com/blog/hypothesis-testing/z-test-in-hypothesis-testing